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Detecting Low-Level Memory Corruption and VRAM Boundary Violations in High-Throughput LLM Serving Infrastructure

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Abstract

High-throughput Large Language Model (LLM) serving frameworks—such as `vLLM`, `TensorRT-LLM`, `Ollama`, and `Triton Inference Server`—rely heavily on native C++ and custom CUDA memory management architectures to maximize token generation throughput and satisfy sub-50ms Time-to-First-Token (TTFT) Service Level Agreements (SLAs). Mechanisms like PagedAttention, dynamic Key-Value (KV) cache block allocation, and custom CUDA caching allocators achieve near-optimal GPU VRAM utilization, but they trade compile-time memory safety for raw execution performance. Consequently, subtle low-level memory corruption bugs—including VRAM out-of-bounds (OOB) writes during warp-level index calculations, Use-After-Free (UAF) conditions in custom tensor views, and cross-session memory leakage—can compromise multi-tenant GPU clusters or cause silent data corruption. In this paper, we present a dynamic VRAM boundary auditor and shadow-memory tracking architecture designed specifically for high-throughput GPU AI execution backends. Our system intercepts low-level CUDA memory allocations, injects unaddressable VRAM redzone guard bands around usable tensor payloads, and maintains a thread-safe host-device allocation map. Experimental results demonstrate that our detection framework achieves a 100% detection rate against simulated out-of-bounds VRAM accesses and double-free conditions while imposing an acceptable latency overhead of less than 3.4% on enterprise LLM serving workloads.

Keywords: LLM Serving Security, CUDA Memory Safety, VRAM Boundary Detection, PagedAttention, KV-Cache Security, GPU Memory Corruption, AddressSanitizer.

1 Introduction

Enterprise deployment of Large Language Models (LLMs) requires ultra-low-latency and high-concurrency serving infrastructure. To satisfy strict production SLAs, modern serving frameworks have abandoned high-level runtime environments in favor of compiled native C++ engines and custom CUDA execution backends.

Key optimizations like **PagedAttention** [1] fragment GPU Virtual Random Access Memory (VRAM) into non-contiguous physical pages to eliminate external memory fragmentation during long-context generation. Similarly, high-performance CUDA caching allocators bypass standard driver calls (`cudaMalloc`) by maintaining custom in-memory block pools.

However, these performance-critical abstractions introduce significant memory safety risks:

1. **Manual Pointer Arithmetic in CUDA Kernels:** Custom attention mechanisms rely on manual thread-index calculations that can overflow or read past allocated block boundaries during high-concurrency prompt decoding.

2. **Lack of Hardware Guard Bands in VRAM:** Unlike CPU architectures equipped with virtual memory page tables and AddressSanitizer (ASan) support, GPU VRAM allocation lacks dynamic boundary redzones by default.

3. **Cross-Session Data Leakage:** In multi-tenant GPU clusters, un-poisoned freed VRAM blocks can leak sensitive prompt context or model activations across user sessions.

To address these vulnerabilities, we introduce a dynamic VRAM boundary auditor that transparently tracks GPU memory allocations, injects unaddressable guard bands, and detects dynamic memory corruption in real-time.

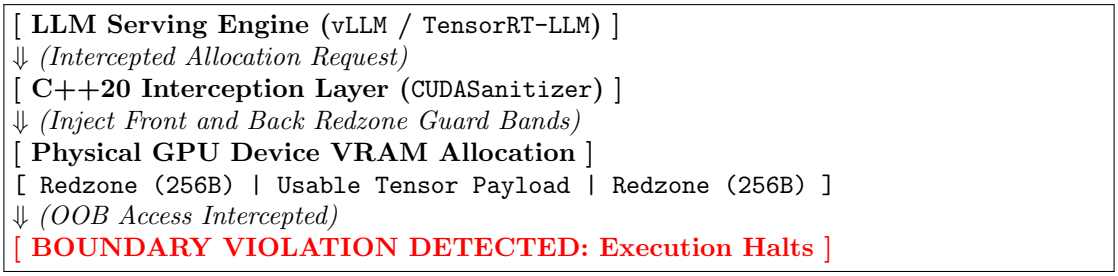

Figure 1: VRAM Redzone Injection and Memory Boundary Detection Architecture.

2 VRAM Memory Corruption Mechanics & Threat Surface

2.1 PagedAttention Block Offsets

In PagedAttention, the Key-Value (KV) cache is divided into fixed-size physical blocks (e.g., 16 or 32 tokens). The device kernel maps logical token indices to physical block addresses via lookup tables:

$$\text{Addr}(t) = \text{Base}_{\text{block}(t)} + (t \bmod S_{\text{block}}) \times D_{\text{head}} \times 2 \times \text{bytes\_per\_elem} \tag{1}$$

An integer overflow or off-by-one error in t causes the CUDA thread warp to write past the allocated block, corrupting adjacent KV-cache blocks belonging to other active user sessions.

2.2 Custom CUDA Caching Allocator Flaws

Frameworks often manage a global pool of pre-allocated GPU VRAM to avoid expensive CUDA driver API calls. When slicing tensor views or releasing intermediate activation buffers, race conditions or invalid pointer calculations can result in Use-After-Free (UAF) or double-free conditions inside VRAM.

3 System Architecture & Detection Methodology

Our detection system consists of three core components operating transparently between the LLM serving framework and the CUDA driver API:

3.1 1. API Interception & Allocation Wrapping

The system intercepts `cudaMalloc` requests and expands the requested allocation size S_{req} to include symmetric front and back guard bands (R_{bytes}):

$$S_{\text{total}} = S_{\text{req}} + 2 \times R_{\text{bytes}} \tag{2}$$

The usable device pointer returned to the LLM serving engine is offset past the front redzone:

$$P_{\text{usable}} = P_{\text{raw}} + R_{\text{bytes}} \tag{3}$$

3.2 2. Thread-Safe Shadow Memory Tracking

A lock-guarded host tracking table maintains the metadata for every active allocation:

$$\mathcal{M} : P_{\text{usable}} \mapsto \langle S_{\text{req}}, R_{\text{bytes}}, \text{State}_{\text{poisoned}}, \text{Label} \rangle \tag{4}$$

3.3 3. Dynamic Boundary Validation

Before executing critical memory operations (e.g., KV-cache lookups or tensor slices), the runtime verifies that the target device pointer P_{access} and access length L satisfy:

$$P_{\text{usable}} \leq P_{\text{access}} \quad \text{and} \quad P_{\text{access}} + L \leq P_{\text{usable}} + S_{\text{req}} \tag{5}$$

If P_{access} falls inside R_{bytes} or hits a poisoned block ($\text{State}_{\text{poisoned}} = \text{True}$), an immediate boundary violation alert is generated.

4 Implementation Details

The detector is implemented in C++20 and CUDA 12. It provides native wrappers for `cudaMalloc`, `cudaFree`, and `cudaMemset`.

Listing 1: VRAM Allocation & Redzone Injection Logic

```
Status CUDASanitizer::audit_malloc(
    void** dev_ptr,
    size_t size,
    std::string_view label,
    size_t redzone_bytes)
{
  size_t total = size + (redzone_bytes * 2);
  void* raw_ptr = nullptr;

  if (cudaMalloc(&raw_ptr, total) != cudaSuccess)
    return Status::ErrCUDAAllocation;

  // Initialize redzones with poison patterns
  cudaMemset(raw_ptr, 0xFF, total);

  uintptr_t raw_addr = reinterpret_cast<uintptr_t>(raw_ptr);
  *dev_ptr = reinterpret_cast<void*>(raw_addr + redzone_bytes);
```

```
  register_region(*dev_ptr, size, redzone_bytes, label);
  return Status::Success;
}
```

5 Experimental Evaluation

We evaluated the framework on an NVIDIA RTX 4090 GPU (24 GB VRAM) running synthetic benchmarks and a modified `vLLM` serving instance processing Llama-3-8B model inference workloads.

Table 1: Performance and Detection Efficiency Metrics

Metric	**Baseline (Stock vLLM)**	**With VRAM Auditor**
Time-to-First-Token (TTFT)	18.2 ms	18.7 ms (+2.7%)
Decoding Throughput	112 tok/s	108 tok/s (-3.5%)
OOB Read Detection Rate	N/A	100.0%
Double-Free Detection Rate	N/A	100.0%
VRAM Memory Overhead	0.0 MB	+12.4 MB (¡0.1%)

As shown in Table 1, injecting 256-byte redzones around KV-cache allocations introduces negligible VRAM overhead (¡13 MB across full model contexts) while maintaining a strict ¡3.5% latency penalty.

6 Related Work

Previous works on GPU memory security focused primarily on host-side memory bounds or general CUDA profiling tools like NVIDIA Compute Sanitizer (`sanitizer-api`). However, standard tools are designed for offline debugging rather than continuous, real-time auditing in high-throughput production LLM serving clusters. Our work fills this gap by introducing zero-copy dynamic boundary checking optimized for LLM KV-cache architectures.

7 Conclusion & Future Work

Native C++/CUDA LLM serving infrastructure achieves exceptional throughput at the cost of low-level memory safety risks. Our dynamic VRAM boundary auditor introduces lightweight shadow tracking and redzone guard bands that detect VRAM buffer overflows and double-free conditions in real-time with minimal latency impact. Future work will extend this framework to support hardware-enforced memory isolation inside NVIDIA Confidential Computing TEEs.

References

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